Architectural Design Justification

Lab 3: Component Modelling & Architectural Pattern Selection
Problem Statement #08 - Digital Campus Library Reservation Gateway

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Architecture Selection

"We chose Layered Architecture for the Digital Campus Library Reservation Gateway system."

Reasoning

Reason 1 - Clear separation of concerns: The system handles three distinct kinds of work: user-facing catalog search and reservation screens (Presentation), core logic such as queue management, fine calculation and hold notifications (Business), and persistent storage of catalog, member and transaction records (Data). A layered structure keeps these concerns isolated, which matches how a small university team is likely to divide development and maintenance work.

Reason 2 - Predictable, consistent data handling: Operations such as placing a hold, computing overdue fines, and returning ISBN search results all depend on a single consistent view of the library database (e.g. a member cannot exceed the maximum hold limit, and fines must reflect the true due date). Routing all business components through one Data layer avoids the data-consistency problems that a distributed, independently-deployed style would introduce for this comparatively small, single-institution system.

Security Advantage

Only the Business layer components (Reservation Queue Manager, Catalog Search, Fine Calculation) are permitted to call the Data layer's Library Database interface; the Presentation layer (Library Portal UI) never accesses the database directly. This layering enforces a single, controlled access path to member records and fine data, so validation rules (e.g. hold limits, authentication checks) cannot be bypassed by a compromised or modified front-end client.

Performance Benefit

Because each layer has a narrow, well-defined interface to the layer below it, the Business layer can cache frequently requested data - such as popular ISBN search results - without the Presentation layer needing any awareness of it. This reduces repeated round-trips to the Library Database for the catalog search operation, which the requirements state must return results for a large catalog with low latency.
